

ML NLP Roadmap

so Natural Language Processing is NLP in which we deal with TEXT type of data, where we convert TEXT data into meaningful VECTORS which is a datatype used to feed into any model.

example:

text $\rightarrow$ vectors

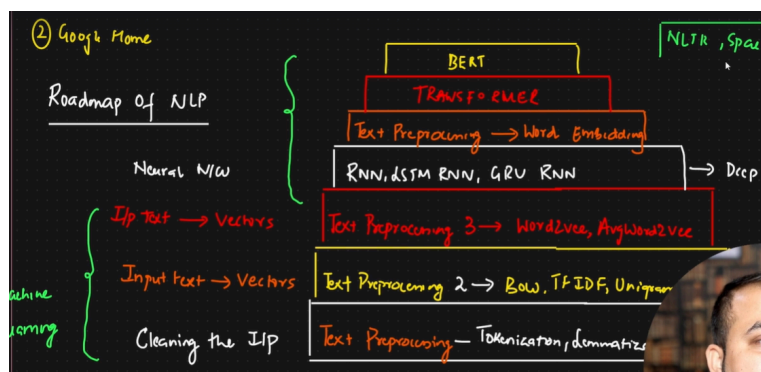
abc $\rightarrow$ [2.8,-0.9,8.7]

this text data needs to be preprocessed in order to convert into a useful token, so preprocessing of texts can be done in many types,

there's a history of the evolution of ways we could preprocess text data, but to understand it in a simpler level,

we will categorize it in a pyramid sort of structure which explains what came first, what is latest and keep in mind, size of model AND accuracy INCREASED as we go UP the pyramid.

this pyramid has the type of text processing and the names of techniques we're going to learn in that type



1 2 3 4 etc is in ascending order of how what came previous in the system and whats now more optimised and uses a heavier model to create super better accuracy

initially

text preprocessing 1 - tokenization (paragraph into sentence, or sentence into word etc), limitization, stemming, stop words, etc

this works towards CLEANING the text

text preprocessing 2 - bag of words, TFIDF, unigrams, bigrams

this works towards CONVERTING the text into vectors

this vector shall be able to capture the context of that sentence

text preprocessing 3 - word2vec, avg word2vec

now moving on to DL models

text preprocessing 4 - RNN, LSTM RNN, GRU RNN

text preprocessing 5 - Word Embedding

text preprocessing 6 - Transformers

text preprocessing 7 - BERT

TOKENIZATION AND BASIC TERMS

terms :

- **Words:** Smallest unit
- **Vocabulary:** Meaningful Words
- **Documents:** Sentences
- **Corpus:** Paragraph

IMPORTANT DIFFERENCE :

Depending on the tokenizer, a token can be:

- A word
- Part of a word
- A character
- A punctuation mark

LLMs do not directly understand words. They first convert text into tokens.

A vector is a numerical representation of something.

Text



Tokens



Vectors (embeddings)



Neural network processing



Output tokens



Output text

Tokenization:

Process of converting either a paragraph or a sentence into TOKENS

if im converting a paragraph into tokens that means that im converting the paragraph into sentences

and if im converting a sentence into tokens that means im converting a sentence into words

meaning you apply tokenization on a particular paragraph and it'll convert into a DOCUMENT,

if you apply tokenization on DOCUMENT it'll break it down into WORDS

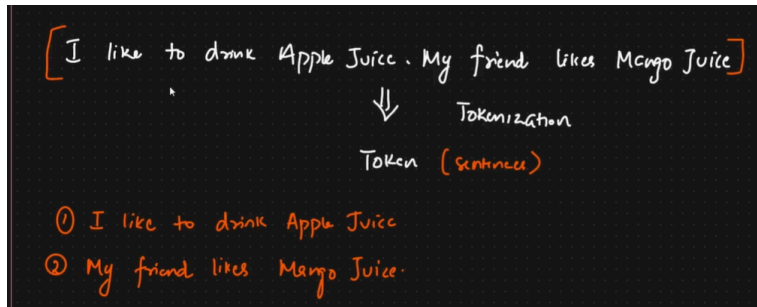
in short WORDS can also be a token

SENTENCES can also be a token

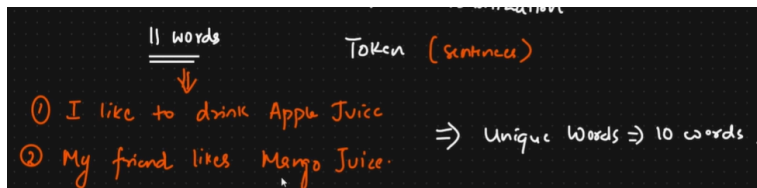
tokenization looks for characters like “.” to see if a sentence has ended

group of sentences (documents) → paragraph (corpus)

group of words → sentence (document)



but unique words are the ones which aren't getting repeated



and vocabulary is nothing but a collection of UNIQUE words

PRACTICALS TOKENIZATION

NLTK is Natural Language Tool Kit (Python library for performing text preprocessing and converting them to vectors)

Feature	NLTK	spaCy
Purpose	NLP education, research, learning concepts	Production grade applications
Speed	Slower	Much faster
Ease of Learning	Great for beginners	Slightly difficult but still straightforward
Pretrained Models	Limited/basic	Powerful pretrained pipelines
Tokenization	Basic	Advanced optimization
POS Tagging	Available	More accurate and faster

Feature	NLTK	spaCy
Named Entity Recognition (NER)	Basic support	Strong NER
Dependency Parsing	Available but less optimized	One of strength
Deep Learning Integration	Not native	Integrated with modern frameworks
Industry Usage	Teaching, research, prototyping	Real-world production applications